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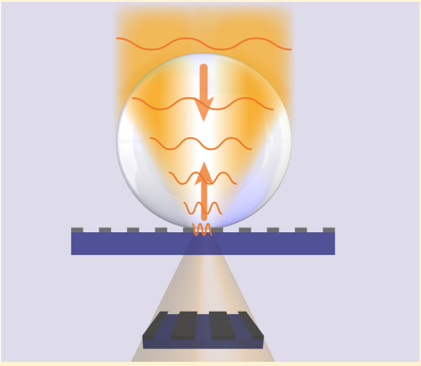
Super-Resolution Imaging of a Dielectric Microsphere Is Governed by the Waist of Its Photonic Nanojet

Hui Yang,[†] Raphaël Trouillon, Gergely Huszka, and Martin A. M. Gijs*

Laboratory of Microsystems, École Polytechnique Fédérale de Lausanne, 1015 Lausanne, Switzerland

Supporting Information

ABSTRACT: Dielectric microspheres with appropriate refractive index can image objects with super-resolution, that is, with a precision well beyond the classical diffraction limit. A microsphere is also known to generate upon illumination a photonic nanojet, which is a scattered beam of light with a high-intensity main lobe and very narrow waist. Here, we report a systematic study of the imaging of water-immersed nanostructures by barium titanate glass microspheres of different size. A numerical study of the light propagation through a microsphere points out the light focusing capability of microspheres of different size and the waist of their photonic nanojet. The former correlates to the magnification factor of the virtual images obtained from linear test nanostructures, the biggest magnification being obtained with microspheres of ∼6−7 μm in size. Analyzing the light intensity distribution of microscopy images allows determining analytically the point spread function of the optical system and thereby quantifies its resolution. We find that the super-resolution imaging of a microsphere is dependent on the waist of its photonic nanojet, the best resolution being obtained with a 6 μm Ø microsphere, which generates the nanojet with the minimum waist. This comparison allows elucidating the super-resolution imaging mechanism.



KEYWORDS: *Microsphere, super-resolution imaging, photonic nanojet, optical microscopy, point spread function*

Conventional optical microscopes are limited by the so-called diffraction limit and can resolve features of around half of the wavelength of illumination λ, as they are only capable of transmitting the propagating wave components emanating from the illuminated object.[1,2] The evanescent components that carry the fine information about the object decay exponentially in a medium with positive permittivity and permeability and are lost before reaching the image plane. Several efforts to this date have been pursued to overcome the diffraction limit using various approaches including near-field scanning probes,[3−6] super/hyper-lenses driven by surface-plasmon excitation,[2,7−12] and fluorescence microscopy with molecular excitation.[13−17] However, the applications of these methods have been limited in part due to their sophisticated engineering designs or pretreatment steps. Alternatively, it has been demonstrated that the use of nanoscale lenses,[18−20] polymer microdroplets,[21] and especially dielectric microspheres[22−30] on top of the objects can achieve near-field focusing and magnification, which in turn results in the capability to resolve features beyond the diffraction limit. Wang et al. first proposed the use of fused silica beads with refractive index $n \sim 1.46$ and diameter from 2 to 9 μm in combination with a conventional optical microscope to achieve a resolution of 50 nm in air with a white light source.[22] Later on, it was reported that large polystyrene microspheres (above 30 μm) can also achieve super-resolution imaging with large field-of-view in air without immersion liquid.[24] Moreover, it was demonstrated that high-index microspheres ($n \sim 1.9$−2.1), fully immersed in liquid, actually allow enhanced imaging with minimum resolved feature sizes of $\sim\lambda/7$ with white-light illumination for imaging of nanofeatures[23] and adenoviruses,[25] as well as with fluorescent microscopic setup for resolving the structures of subcellular organelles.[26] More recently, microspheres were used in combination with confocal microscopy for achieving a 25 nm lateral resolution under 408 nm wavelength illumination.[27] In these works, the microspheres are placed on top of the sample object, where they collect the underlying sample's near-field nanofeatures and subsequently transform the near-field evanescent waves into far-field propagating waves, creating a magnified image in the far-field, which is collected by a conventional optical microscope.[22,26,30] The super-resolution capability of microspheres is linked to two factors: (i) the microsphere performs as a solid-immersion lens and provides local enhancement of the refractive index and a reduction of the illumination wavelength; (ii) the development of the photonic nanojet.[26,30] A substantial literature has developed regarding the existence, properties, and potential applications of the photonic nanojet.[31−56] In principle, a photonic nanojet is a narrow light beam with high optical intensity that can be generated by a transparent dielectric symmetric body, like a microcylinders or microsphere, upon illumination. The nanojet that emerges from a microsphere locates in the immediate

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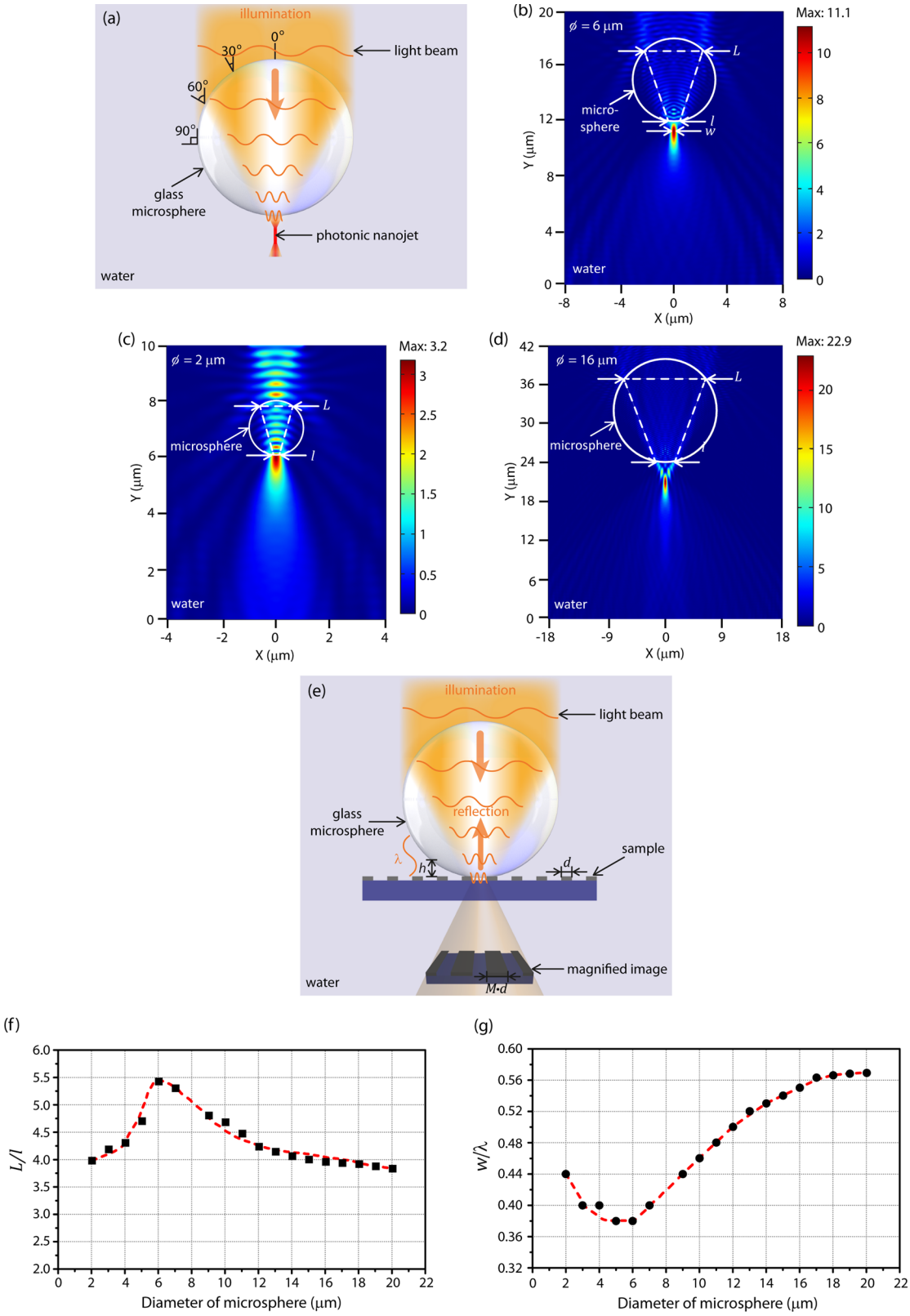

Figure 1. Mechanism of nanojet generation and imaging of a dielectric microsphere. (a) Focusing of a plane wave light beam into a nanojet at the rear surface of a free-standing microsphere. At the front surface of the microsphere, the light is refracted at low incidence angle, while at higher incidence angle it gets mostly reflected. (b) FEM simulation of the light propagation through a 6 μm Ø barium titanate microsphere in water. The linear region where substantial refracted light enters the microsphere at its front surface is referred to as L, while the width of the exiting beam at the rear surface is denoted as l; the waist of the nanojet is referred to as w. (c,d) FEM simulation of the light propagation through 2 and 16 μm Ø barium titanate microspheres in water medium, respectively. (e) When a microsphere is positioned on a grating structure with line width d and illuminated from the front, the light reflected by the grating allows detecting a virtual image with magnification factor M. When the distance h between the microsphere and the grating is small enough (of order of the illumination wavelength λ), the near-field evanescent wave carrying the fine details of

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Figure 1. continued

the grating can become propagating in the high refractive index sphere, and later in the medium where it is to be collected by the microscope objective. (f) FEM simulation results of the light focusing capability of a microsphere, expressed by the ratio L/l, as a function of the microsphere diameter. The dots are obtained from the simulation, while the red dotted line is a guide to the eye. (g) FEM simulation results of the normalized waist of the photonic nanojet w/λ, as a function of the microsphere diameter. The dots are obtained from the simulation, while the red dotted line is a guide to the eye.

vicinity of the rear-surface of the sphere. It is a nonresonant phenomenon that appears for spheres with a diameter of the order of 10−100 λ and with a ratio of their refractive index n_{ms} to the refractive index of the background medium (water in this study) n_w that is smaller than about 2. The nanojet can maintain a subwavelength full width at half-maximum (fwhm) transverse beam width along a path that can extend more than ∼2λ beyond the sphere, and the minimum fwhm beam width, referred to as "waist" in this paper, can be smaller than the classical diffraction limit, in fact as small as ∼ $\lambda/3$.

The imaging resolution of a classical microscope depends on the size of the spot that is generated by focusing the incident propagating wave in the far-field and is therefore limited by diffraction. For microsphere-assisted optical microscopy, evanescent waves close to the surface of the microsphere play a significant role. Even though it was suggested that the development of the photonic nanojet is essential to the super-resolution imaging capability of a microsphere,[26,30] the exact link remained unclear. Here we study, in a quantitative way the role of the photonic nanojet for super-resolution imaging. First a systematic numerical study of the light propagation through microspheres of different size using the finite element method (FEM) is performed. This allows characterizing the photonic nanojet at the rear-surface of a microsphere and relating the microsphere's theoretical magnification factor to the light focusing capability of the photonic nanojet. Second, we perform an experimental study in which a systematic series of barium titanate glass microspheres with diameter from 3 to 21 μm are used to image linear test nanostructures that are immersed in water. The experimental magnification factor and the point spread function that is analytically determined from the images allow evaluating the resolution of the optical system, which is shown to directly correlate with the calculated properties of a microsphere's photonic nanojet.

As schematically illustrated in Figure 1a, when a microsphere is illuminated by a propagating light beam from the far-field, the light is mostly refracted on its frontal surface at small incident angle and reflected at higher incident angle, the limiting angle between the two regions given approximately by the Brewster angle of the optical interface. When the size of the microsphere is much bigger than the illumination wavelength, it is a good approximation to use ray optics to explain how the light that is incident on the top-surface of the microsphere propagates. The refracted light propagates through the microsphere and generates a photonic nanojet near the rear-surface of the microsphere (details described in the Supporting Information). An FEM study of the light wave propagation through a barium titanate glass microsphere (n_{ms} = 1.92), with diameter Ø ranging from 2 to 20 μm in surrounding water medium (n_w = 1.33) is performed (details described in Methods). An electromagnetic wave with wavelength λ = 600 nm is applied to a boundary with a length that is the same as the size of the microsphere and that is far away from its top-surface. As the size of the boundary is much longer than the wavelength, a plane-wave incident light beam is a good approximation for studying the photonic nanojet. Figure 1b−d shows the light intensity distributions near a 6, 2, and 16 μm microsphere, respectively. When the microsphere is 6 μm in size (Figure 1b), the photonic nanojet directly emerges from the microsphere surface. When decreasing the microsphere's size, the focal point of the nanojet moves into the microsphere, as seen for the simulation result of a 2 μm Ø microsphere (Figure 1c). For a microsphere size bigger than 6 μm, the external focal point of the nanojet moves away from the rear-surface of the microsphere and Figure 1d shows the simulation result obtained for a 16 μm Ø microsphere. To quantitatively study how light is focused by the microsphere into the photonic nanojet, we determined from the model calculation the linear region where substantial refracted light enters the microsphere at its front surface, referred to as L in Figure 1b−d, while the width of the exiting beam at the rear surface is denoted as l. The waist of the nanojet is referred to as w (see Figure 1b), which is the fwhm of the nanojet along the x-axis at the peak intensity of the y-axis. It should be noted that the light source in the FEM study is actually coherent, while the light source in conventional optical microscopy is incoherent. The comparison on the light intensity distribution in the photonic nanojet under coherent and incoherent illumination is further discussed in the Supporting Information. The study shows that the intensity profile along the nanojet waist does not significantly change, when using either an incoherent or a coherent light source, indicating that a coherent light-based simulation can provide sufficient information to study the resolution of the imaging system.

The imaging mechanism of a dielectric microsphere is schematically illustrated in Figure 1e. When the focused light that exits the microsphere illuminates the sample object (in this case a linear grating structure with feature size d), no nanojet is generated but instead the reflected light follows a reflection-symmetric optical path, while evanescent waves that contain the high spatial-frequency information on the object are converted into propagating waves within the microsphere (see Supporting Information) when the distance h between the microsphere and the grating is small enough (of order of the illumination wavelength λ). In the meanwhile, a magnified virtual image is generated in the far-field with magnification factor M. The imaging capability of the microsphere directly correlates with the formation of the focused photonic nanojet as obtained from the numerical study, as the same optical paths are involved. Figure 1f,g shows the calculated light focusing capability of a microsphere, expressed by the ratio L/l, and the waist of the photonic nanojet normalized by the illumination wavelength w/λ as a function of the microsphere diameter, respectively. A bigger L/l indicates a better light focusing by the microsphere and corresponds to a smaller waist of the nanojet. According to our simulations, the microsphere with diameter of 6 μm shows the best light focusing capability and smallest waist of the photonic nanojet.

To experimentally study the super-resolution imaging effect, grating structures consisting of 120 nm wide lines with 100 nm

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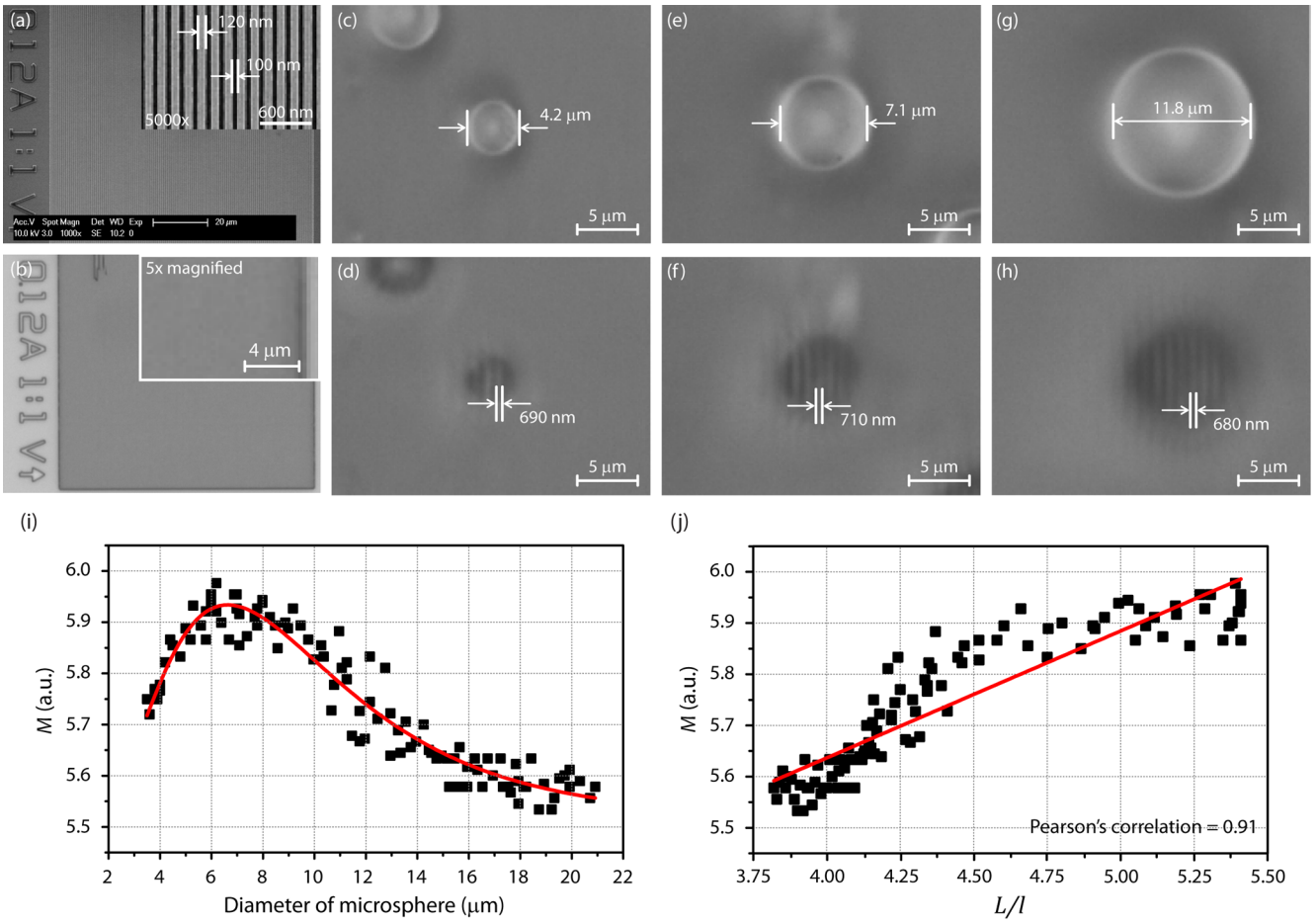

Figure 2. Imaging of a grating nanostructure using different size microspheres and a water-immersion objective. (a) Scanning electron microscopy image of the silicon grating test structure containing 120 nm wide lines with an interspacing of 100 nm. (b) Optical microscopy image of the nanostructure taken by a 40× water-immersion objective with NA of 0.8. The insert is a 5× magnified image, clearly showing that conventional microscopy cannot resolve the nanostructures with this feature size. (c−h) Optical microscopy images obtained by positioning on the grating microspheres with sizes of (c,d) 4.2 μm, (e,f) 7.1 μm, and (g,h) 11.8 μm, respectively. The images of (c,e,g) are focused on the microspheres' center plane, while the corresponding images (d,f,h) are focused on the virtual image plane, showing that the grating nanostructure is imaged with a different magnification factor M for microspheres of different sizes. (i) The dots represent the experimental magnification factor as a function of the microsphere diameter, while the solid line is a guide to the eye. (j) The experimental magnification factor M as a function of the light focusing capability L/l obtained from the simulations. The solid line represents a linear fitting curve with a Pearson's correlation coefficient of 0.91.

interspacing (see Figure 2a) immersed in water are used as test samples to be imaged with transparent barium titanate glass microspheres that are loosely positioned on top of them. A conventional reflection microscope (Zeiss Axioplan microscope) equipped with a CCD camera (AxioCam MRm camera) and a 40× water immersion objective with numerical aperture (NA) of 0.8 is used to take images. A halogen lamp is used as the white-light illumination source with a wide-band spectrum, which is from ∼400 to 700 nm and the peak appears at $\sim\lambda$ = 600 nm (more details are shown in the Supporting Information). Figure 2b is a microcopy image of the gratings taken by the 40× objective along, the inset is a 5× magnified image, clearly showing that the conventional optical microscope cannot resolve the nanopatterns with a feature size of ∼100 nm. However, when a microsphere is placed on top of the nano-objects, the subdiffraction features become observable. Figure 2c,e,g shows microspheres with diameters of 4.2 μm, 7.1, and 11.8 μm positioned on the sample, respectively. The focal plane of the microscope coincides with the plane of maximum sphere diameter, hence, on these pictures, the grating nanostructures are out-of-focus. When the focal plane of the microscope is set to the image plane of the microsphere, images obtained with the microspheres of Figure 2c,e,g are shown in Figure 2d,f,h, respectively. Compared to the size of the objects, the feature sizes of the line structures in the images are clearly magnified. As already schematically illustrated in Figure 1e, comparing the size of the object with that of the image allows determining the magnification factor M. The experimental M values versus the microsphere diameter are plotted in Figure 2i. We can see that the biggest magnification is obtained with microspheres of ∼6−7 μm in size. When we compare the experimental magnification factor M with the theoretical light focusing capability L/l, as obtained from the simulations, a positive correlation with a Pearson's correlation coefficient of 0.91 is obtained, indicating that a better light focusing capability of a microsphere logically results in a higher magnification factor.

A remaining issue is whether the light focusing capability of a microsphere has any impact on its super resolution imaging. For solving this question, linear test nanostructures with 300 nm wide lines and 900 nm interspacing are imaged through the microspheres with different size by using the same optical microscope setup. As illustrated in Figure 3a, the microsphere is

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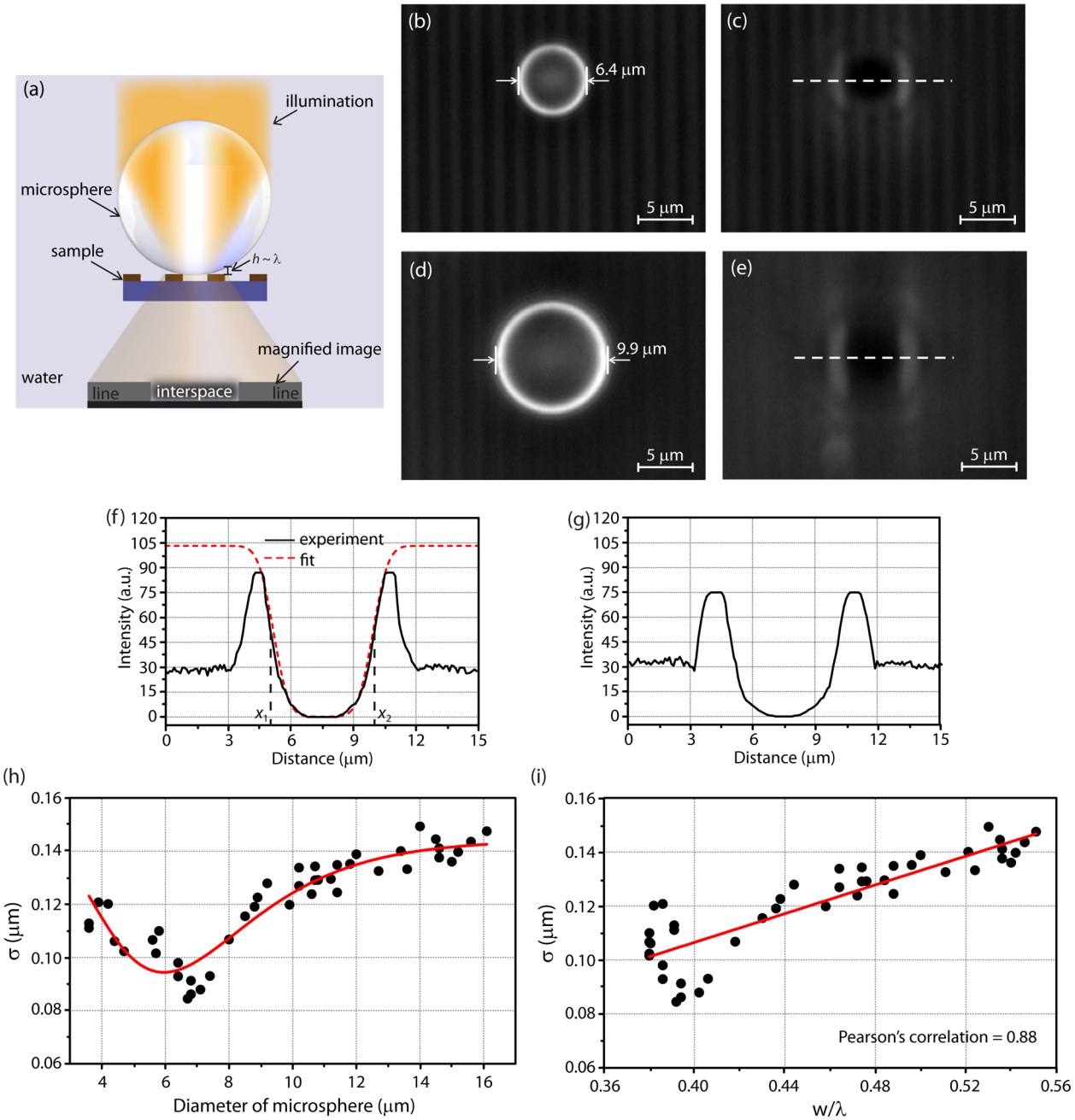

Figure 3. Quantification of the experimental resolution using the analytical point spread function model and correlation with the waist of the photonic nanojet. (a) A microsphere is positioned specifically in the middle of two lines of a dedicated test grating (line width of 300 nm and interspacing of 900 nm) to characterize the sharpness of line/interspacing boundary in the virtual image. (b−e) Optical microscopy images obtained by positioning on the grating microspheres with sizes of (b,c) 6.4 and (d,e) 9.9 μm, respectively. The images of (b,d) are focused on the microspheres' center plane, while the corresponding images (c,e) are focused on the virtual image plane. Each dashed line indicates where the intensity profile will be taken that is to be fitted with the analytical point spread function model. (f) Intensity distribution along the dashed line in panel c with x_1 and x_2 the positions of the descending and ascending steps obtained from the fit using eq 5. (g) Intensity distribution along the dashed line in panel e. (h) The actual image standard deviation σ that is obtained from the fit, related to the true resolution of the system, as a function of microsphere size. The dots are obtained from the fits with the analytical model, while the solid curve is a guide to the eye. (i) The correlation between σ and the normalized waist of the photonic nanojet w/λ. The solid line represents a linear fitting curve with a Pearson's correlation coefficient of 0.88.

positioned in the middle of two lines and the magnified image can be obtained when the distance h between the microsphere and the grating is of order of the illumination wavelength λ (see Figure 1e), therefore only the interspacing at the center of the microsphere (darker zone in the image) and its neighboring two lines (clearer zones in the image) are visible in the far-field. Figure 3b,c shows the microscopic images obtained with the 40× water-immersion objective (NA = 0.8) focusing on a 6.4 μm microsphere and on the image plane, respectively. The light intensity profile along the dashed line in Figure 3c is shown in Figure 3f. Moreover, similarly taken images obtained from a 9.9 μm microsphere are shown in Figure 3d,e, while the light

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intensity profile along the dashed line in Figure 3e is shown in Figure 3g. Comparing Figure 3f with Figure 3g, clear differences on the peak intensities and on the intensity profiles are observed.

This type of image is further analyzed using an analytical point spread function (PSF) model to study the resolution in a quantitative way. The PSF is an important factor characterizing the mechanisms underlying the formation of an image in an optical system, defined as the response of the imaging system to a point object.[57−59] This three-dimensional function is characteristic of the imaging system and can have a rather complicated analytical expression, hence sometimes requiring simplifications or approximations to facilitate its use. However, the shape of the PSF is directly related to the resolution of the system, as the narrower the PSF, the better the resolution. The imaging process described here is based on the collimation of light that is incident on top of the microsphere into the nanojet. For detection of an object, the light is reflected back from the object into the detector, essentially following the same mirror-symmetric trajectory as during incidence. The width of the nanojet l and more precisely the quantity L/l detailed in Figure 1b is a measure of the focusing capability of a microsphere. Indeed, the smaller l and the larger L/l is, the easier it will be for an initially evanescent wave, when it is reflected from the object, to be converted to a wave with higher spatial frequency that becomes propagating inside the microsphere. If higher spatial frequencies can be detected, this will result in a sharper image, hence in a better resolution and a narrower PSF. That is why the theoretical simulations of the nanojet profiles, provided, for instance, in Figure 1, are indicative of the diffraction processes in the optical system and can be useful to compare with the experimental PSF. Additionally, simulations hint that the illumination profile partially describes the blurring introduced by the system and is at least indicative of the PSF in the absence of severe mismatches in the refractive indices.[60] As a consequence, these illumination profiles are used below to establish assumptions on the overall PSF, and are justified a posteriori.

Typically, the PSF can be approximated through models and numerical simulations. However, the purpose of this section is to find an experimental approach to evaluate the lateral resolution of the system through the PSF and to compare it to some of the results predicted by the simulations. In the case considered in this work, that is, where a plane is imaged, only the $x-y$ profile of the PSF at the waist of the nanojet is relevant to the resulting image. Moreover, because of the cylindrical geometry of the problem, as shown in Figure 1b−d, only the expression of the radial component at the waist (that is, along the x-axis in Figure S2c) is required to describe the PSF. Furthermore, only sections of the images located at the center of the microsphere are analyzed. This guarantees that distortions due to the spherical shape of the microlens are limited, and that small translations along the x- and y-axes at the vicinity of the central axis of the system do not dramatically alter the image. This is supported experimentally by the images shown in Figure 2d,f,h where the gratings are clearly resolved at the center of the microsphere and are parallel, hence ensuring that no dramatic radial distortion occurs. As a consequence, at the center of the microsphere the imaging device can be assumed to be largely shift independent in the $x-y$ plane. Combined with the linearity of the system, this fact suggests that the image can be expected to be solely determined by the intensity profile of the PSF.[61] Furthermore, as shown by the shape of the illumination profiles discussed in the Section S6 of the Supporting Information, the shape of the nanojet is independent of phase shifts. As the final image is formed by the illumination light reflected by the object, one can assume that the whole system and therefore the PSF are not dependent on phase shifts. Briefly, it allows to relate the recorded two-dimensional (2D) image Im to the input object Ob through a convolution operation (denoted as *)[62,63]

$$\mathrm{Im}(x, y) \equiv (\mathrm{PSF}*\mathrm{Ob})(x, y) \tag{1}$$

The PSF can be defined as the impulse response function of the system, that is, the image obtained from the imaging of a point. Mathematically, this impulse, or point, can be expressed as the Dirac function δ, the function returning 0 for $x \neq 0$ and $y \neq 0$, and whose integral is 1 over $\mathbb{R}^2$. More intuitively, δ can be approximated as an infinitely high, infinitely sharp peak centered over (0, 0), the function being equal to 0 anywhere else. Furthermore, δ is the unit element for convolution, hence

$$\mathrm{PSF}(x, y) = (\mathrm{PSF}*\delta)(x, y) \tag{2}$$

Direct measurement of the PSF can be challenging, as obtaining a pure point as the object to image is impossible. It can be approximated by imaging a very small disk for instance, but the result can be distorted if the object is below the imaging capabilities of the device. A numerical simulation, as shown in Figure 1, or an exact analytical solution can be used, but this is not always available. Moreover, the purpose of this analysis is to confirm the results of the numerical simulation (Figure 1f,g) with an experimental approach. To experimentally characterize the PSF, it has been suggested to image a step, corresponding to a Heavyside function H along one of the two dimensions of the image, here, for instance, along the x-axis. This is a more rigorous and elegant way to evaluate the PSF and also the reason why the grating nanostructure is used as sample in this work. Indeed, δ is the derivative of H along the x-axis,[64] and the convolution operation is stable through differentiation, leading to

$$\mathrm{PSF}(x) = \frac{\partial}{\partial x}(\mathrm{PSF}*H)(x) \tag{3}$$

It is commonly assumed that the PSF is accurately described with a 2D Gaussian,[65−71] which can be reduced to one-dimension (1D) in the case of imaging a step along the x-axis. Even though an Airy function can be considered, as it describes the diffraction pattern generated by a small circular aperture on the xy-plane, numerical investigations show that a Gaussian is also a very common fit in the xy-plane for the PSF of a confocal microscope.[72] To confirm the validity of using a Gaussian approximation in contrast to an Airy approximation, the intensity profiles at the waist along the x-axis for the simulations of the nanojet profiles shown in Figure 1b−d were fit with a Gaussian and an Airy function. In both cases, these fittings were found to be imperfect (Figure S5) but nevertheless resulted in comparably good fits ($R^2 > 0.966$). As the use of a Gaussian facilitates considerably the calculations and the numerical analysis, this function was chosen to approximate the overall PSF of the system in the rest of this study.

Thanks to this approximation, the image can now be obtained by convoluting the step profile (the object) with a 1D Gaussian (the PSF) characterized by a standard deviation σ (see eq S4). By integrating eq 3 along the x-axis, the image

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actually results in the integral of a Gaussian (the PSF) and is by definition described by the error function erf

$$\mathrm{erf}(x) \equiv \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} \mathrm{d}t \tag{4}$$

By fitting the profile of the image with erf, we can extract the σ associated with the PSF, which is directly associated with the lateral resolution of the imaging system.[73]

Experimentally, the microsphere sits in the middle of two lines. As shown in Figure 3c,e, two stripes are observed for each microsphere, resulting in two opposite steps. In agreement with eq 4, the obtained image profile should be described by the function f

$$f(x) = \alpha\left(\mathrm{erfc}\left(\frac{(x - x_1)}{\sigma'\sqrt{2}}\right) + \mathrm{erf}\left(\frac{(x - x_2)}{\sigma'\sqrt{2}}\right)\right) + \text{constant} \tag{5}$$

where erfc is the complementary error function, α is a constant and x_1 and x_2 are the positions of the descending and ascending steps, respectively (shown in Figure 3f), which together define the magnification of the image. The image standard deviation σ' is obtained from the fitting, and has to be divided by the amplification factor associated with the microsphere size (as shown in Figure 2i) to obtained the actual σ. A description of a typical experimental image, along with the fitted f, is shown on Figure 3f. From this fitting, the characteristic σ for each microsphere size can be computed and is presented on Figure 3h. The other fitting parameters are discussed in the Supporting Information and support the analysis presented here with $x_2 - x_1 \sim 900$ nm, which is in good agreement with the geometry of the sample, and $\alpha \sim 100$ for all the beads considered. In particular, the fact that the distance $x_2 - x_1$ was correctly evaluated further validates the shift invariance assumption. If this was not the case, as this parameter is measured over a large part of the field of view appearing on the microsphere, a significant distortion would have occurred thus preventing and accurate measurement of the intergrating distance. The best σ is obtained with 6 μm size microspheres and is below 100 nm. The experimental σ is also in good agreement with the nanojet waist values derived from the simulations. The actual σ fitted from the experimental data as a function of w/λ obtained from the simulations is plotted in Figure 3i: a positive correlation with the Pearson's efficiency of 0.88 is obtained, indicating that the imaging resolution of the microsphere is clearly dependent on the waist of the photonic nanojet. Additionally, w is defined at the fwhm of the nanojet along x, which in a Gaussian approximation is $2\sqrt{2\ln(2)}\,\sigma \approx 2.355\sigma$. As the simulated w is ~240 nm (for $\lambda = 600$ nm) and the experimental σ is ~100 nm, the correspondence between the simulation and the experimental analysis is excellent. This good agreement validates a posteriori the assumptions on the overall PSF shape drawn from considering the simulated nanojet profiles. It also supports the validity of the simulations, and that the resolution of the system is largely controlled by the illumination pattern, that is, the width of the waist of the nanojet.

Considering that the illumination source has a spectral range from ~400 to 700 nm (see Supporting Information) and a 100 nm structure is resolved in the experiments, the resolution res is therefore in between $\lambda/4$ and $\lambda/7$ with λ the illumination wavelength in vacuum. According to the Rayleigh criterion, the minimum feature size that can be resolved by a water-immersion objective with NA = 0.8 is 188 nm under an illumination wavelength of 400 nm. The 100 nm resolution is therefore obtained thanks to the use of microspheres, which convert the evanescent waves with the high spatial-frequency information on the object into propagating waves within the microsphere (see Supporting Information).

Moreover, the minimum waist of the photonic nanojet is ~0.4 λ (Figure 3i). The relationship between the resolution and the nanojet waist can hence be written as res $\sim 0.36w - 0.63w$. The diffraction limit of an objective or lens is defined as $\lambda/(2\mathrm{NA})$. Our method can resolve an object with feature size of 100 nm under a wide-band illumination with spectrum from 400 to 700 nm, so that the NA of the imaging system is obtained as 2−3.5 when considering the minimum and maximum illumination wavelength, respectively. This means that in our study the use of the 40× microscope objective together with a 6 μm Ø microsphere would permit a resolution that would be provided by a hypothetical (as nonexisting) microscope objective with NA = 2−3.5.

In conclusion, we reported imaging of a sample's nanofeatures beyond the classical diffraction limit by using a conventional optical microscope in combination with a series of barium titanate glass dielectric microspheres of different sizes. A FEM study on light propagation revealed the light focusing capability of a microsphere of a given size and the generation of the photonic nanojet. By comparing the experimental imaging results with the numerical study, we found that the magnification factor obtained from the virtual images is highly correlated to the calculated light focusing capability of a microsphere. Moreover, we quantitatively studied the resolution of the microspheres of different sizes by analyzing the images and fitting the results with a mathematical model based on the PSF. Our work demonstrated the intimate link of the super-resolution imaging mechanism of a dielectric microsphere with its light focusing capability and the development of the photonic nanojet. Indeed, the combination of refractive and interferometric effects of the incident light produce the narrow-waist photonic nanojet that exits the microsphere. In the imaging mode of the microsphere, identical optical paths are used for generating the magnified image and, therefore, the degree of focusing of the incident light into a nanojet is closely related to the possibility of a microsphere to transform a high spatial frequency evanescent wave generated by the object into a propagating wave that becomes detectable in the far field. We believe that due to these physical insights, dielectric microspheres will be increasingly used in the future, providing a straightforward and robust tool to be integrated with a conventional microscope for super-resolution optical microscopy. This will allow affordable super-resolution imaging of a whole range of samples and biological objects, such as virus particles, labeled nucleic acids and molecules.

Methods. *Numerical Simulation.* The numerical study on light propagation through the microspheres and surrounding water medium is carried out by FEM in COMSOL Multiphysics software. A scalar equation is used to study transverse electric waves in a 2D model. A light source with wavelength of 600 nm and the same width of the microsphere size is set away from the front-surface of the microspheres, 600 nm corresponding to the peak of the halogen lamp that is used as the white-light illumination source in the experiments. After meshing of the model, the element size is ~22 nm, that is, 1/30 of the wavelength, which is sufficiently small to obtain a precise solution. Microspheres with a size ranging from 2 to 20 μm are analyzed in individual models. After each model is solved, L is

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obtained by measuring the distance between the two points with maximum light intensity on the front-surface of the microsphere incident with the illumination light, l is obtained by measuring the distance between the two points with maximum light intensity on the rear-surface where light exits the microsphere, and w is the fwhm of the nanojet along the x-axis at the peak intensity of the y-axis.

Experimental Section. The silicon grating structures consisting of 120 nm wide lines with 100 nm interspacing (in Figure 2) are on a MetroChip microscope calibration target, which is obtained from Pelco (Redding, CA, U.S.A.). The grating structure with 300 nm wide lines and 900 nm interspacing (in Figure 3) is made by chromium on a glass substrate by photolithography. The optical microscopic images are obtained by using a Zeiss Axioplan microscope mounted with a AxioCam MRm camera (Carl Zeiss GmbH, Oberkochen, Germany) and a 40× water immersion objective with NA of 0.8. A halogen lamp is used as the white-light illumination source with a peak at λ = 600 nm.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.nanolett.6b01255.

Additional information, figures, and table(PDF)

■ AUTHOR INFORMATION

Corresponding Author

*E-mail: martin.gijs@epfl.ch.

Present Address

†(H.Y.) IMEC, Leuven, Belgium

Notes

The authors declare no competing financial interest.

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